

# MED.MIX OS

Business Management Operating System



Version 1.0 — June 2026  
Saudi Arabia Market — Arabic-First CRM

Deployed on Railway + Vercel  
Frontend: <https://frontend-steel-three-58.vercel.app>  
Backend: <https://backend-production-40db.up.railway.app>

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# 1. Introduction

MED.MIX OS is a comprehensive business management operating system designed specifically for the Saudi Arabian market. It provides an all-in-one platform for managing customers (CRM), sales leads, quotations, tasks, and business analytics with full Arabic (RTL) support.

The system is built as a modern cloud-native application using FastAPI for the backend and Next.js for the frontend, deployed on Railway and Vercel respectively. It features real-time multi-user synchronization via WebSockets, an AI-powered assistant using OpenAI, and business data import from OpenStreetMap.

## Key Features

- **Arabic-First Interface:** Full RTL support with Arabic labels, dates, and number formatting
- **CRM:** Manage companies, contacts, and projects with full CRUD operations
- **Lead Management:** Track sales leads through pipeline stages (New → Qualified → Contacted → Negotiating → Won/Lost)
- **Quotations:** Create and manage price quotes for clients
- **AI Assistant:** Context-aware business assistant powered by OpenAI GPT-4o-mini
- **Real-Time Sync:** WebSocket-based instant updates across all connected users
- **Business Search:** Find and import companies from OpenStreetMap (free, no API key)
- **Analytics:** Visual dashboard with lead distribution, activity feed, and KPIs
- **Role-Based Access:** JWT authentication with role-based permissions
- **Task Management:** Track and assign tasks within the organization

## 2. System Architecture

MED.MIX OS follows a modern three-tier architecture with clear separation of concerns:

### Architecture Overview

#### Frontend (Presentation Layer)

Next.js 14 (App Router) hosted on Vercel  
→ RTL Arabic UI with Tailwind CSS  
→ Server-side rendering for performance  
→ Client components for interactivity

#### Backend (API Layer)

FastAPI application hosted on Railway  
→ RESTful API endpoints  
→ WebSocket endpoint for real-time  
→ SQLAlchemy async ORM  
→ JWT authentication middleware

#### Database (Data Layer)

PostgreSQL on Railway  
→ Relational schema with 10+ tables  
→ Async queries via asyncpg driver  
→ Redis for caching (optional)

#### External Services

OpenAI API → AI Assistant  
OpenStreetMap → Business search  
SMTP (optional) → Email notifications

### Data Flow

1. User interacts with the Next.js frontend (Arabic RTL UI)
2. Frontend sends API requests to FastAPI backend (with JWT token)
3. Backend validates auth, processes request, queries PostgreSQL
4. Response returned as JSON to frontend
5. On data change, backend broadcasts via WebSocket to all connected clients
6. All clients automatically refresh their UI

# 3. Technology Stack

| Layer    | Technology             | Version | Purpose                         |
|----------|------------------------|---------|---------------------------------|
| Frontend | Next.js                | 14.x    | React framework with App Router |
|          | Tailwind CSS           | 3.x     | Utility-first styling           |
|          | TypeScript             | 5.x     | Type-safe JavaScript            |
|          | Lucide Icons           | latest  | Icon library                    |
| Backend  | Python                 | 3.12    | Runtime                         |
|          | FastAPI                | 0.115.x | Async web framework             |
|          | SQLAlchemy             | 2.0.x   | Async ORM                       |
|          | Pydantic               | 2.x     | Data validation                 |
|          | python-jose            | latest  | JWT tokens                      |
|          | passlib                | latest  | Password hashing                |
|          | OpenAI SDK             | 1.50.x  | AI assistant                    |
|          | httpx                  | latest  | Async HTTP client               |
|          |                        |         |                                 |
| Database | PostgreSQL             | 16.x    | Primary database                |
|          | asyncpg                | 0.30.x  | Async PG driver                 |
| Infra    | Railway                | -       | Backend hosting                 |
|          | Vercel                 | -       | Frontend hosting                |
|          | GitHub                 | -       | Source control                  |
| AI       | OpenAI GPT-4o-mini     | -       | AI assistant model              |
| Maps     | OpenStreetMap Overpass | -       | Business data API               |

## 4. Quick Start & Deployment

### Local Development

```
# 1. Clone the repository
git clone https://github.com/Fly89by/medmix.git
cd medmix

# 2. Backend setup
cd apps/backend
python -m venv .venv
.venv\Scripts\activate # Windows
pip install -r requirements.txt

# 3. Configuration
copy .env.example .env # Edit with your settings

# 4. Run with Docker (recommended)
cd infrastructure/docker
docker compose -p medmix up -d

# 5. Frontend setup
cd apps/frontend
npm install
npm run dev
```

### Deploy to Railway + Vercel

- **Railway (Backend):** Connect GitHub repo → Set root as project → Add PostgreSQL → Set environment variables → Deploy → Set domain target port to 8000
- **Vercel (Frontend):** Import GitHub repo → Set Root Directory to apps/frontend → Add NEXT\_PUBLIC\_API\_URL env var → Deploy
- **Post-Deploy:** Create admin user via seed\_admin.py script or registration API

### Seed Admin User

```
# Run from apps/backend/
python seed_admin.py
# Creates admin@medmix.com / admin123
```

# 5. Authentication & Authorization

## Authentication Flow

MED.MIX OS uses JWT (JSON Web Tokens) for stateless authentication. Tokens are issued at login and must be included in the Authorization header for all protected endpoints.

## Endpoints

| Endpoint           | Method | Description              |
|--------------------|--------|--------------------------|
| /api/auth/register | POST   | Create new user account  |
| /api/auth/login    | POST   | Login, returns JWT token |
| /api/auth/me       | GET    | Get current user profile |

## Token Usage

```
# Include in all API requests
Authorization: Bearer <your-jwt-token>
```

## Roles

| Role    | Permissions                                                |
|---------|------------------------------------------------------------|
| admin   | Full access to all features and settings                   |
| manager | CRUD on companies, contacts, leads, quotes; view analytics |
| user    | Basic CRUD on assigned records; view own tasks             |

# 6. API Reference

Base URL: `https://backend-production-40db.up.railway.app/api`  
All endpoints require JWT authentication unless marked as public.

## Dashboard

| Endpoint           | Description                                  |
|--------------------|----------------------------------------------|
| GET /api/dashboard | Get KPIs, lead distribution, recent activity |

## CRM (Companies, Contacts, Projects)

| Endpoint                   | Description                |
|----------------------------|----------------------------|
| GET /api/companies         | List companies (paginated) |
| POST /api/companies        | Create company             |
| GET /api/companies/{id}    | Get company details        |
| PUT /api/companies/{id}    | Update company             |
| DELETE /api/companies/{id} | Delete company             |
| GET /api/contacts          | List contacts              |
| POST /api/contacts         | Create contact             |
| GET /api/projects          | List projects              |
| POST /api/projects         | Create project             |

## Leads

| Endpoint                      | Description                        |
|-------------------------------|------------------------------------|
| GET /api/leads                | List leads (paginated, filterable) |
| POST /api/leads               | Create lead                        |
| GET /api/leads/{id}           | Get lead details                   |
| PUT /api/leads/{id}           | Update lead                        |
| DELETE /api/leads/{id}        | Delete lead                        |
| POST /api/leads/import/search | Search & import from OpenStreetMap |

## Quotes

| Endpoint             | Description       |
|----------------------|-------------------|
| GET /api/quotes      | List quotations   |
| POST /api/quotes     | Create quotation  |
| GET /api/quotes/{id} | Get quote details |



|                         |              |
|-------------------------|--------------|
| PUT /api/quotes/{id}    | Update quote |
| DELETE /api/quotes/{id} | Delete quote |

## AI Assistant

| Endpoint                   | Description                  |
|----------------------------|------------------------------|
| POST /api/assistant/chat   | Send message to AI assistant |
| GET /api/assistant/context | Get CRM context for AI       |

## Knowledge Base

| Endpoint                   | Description            |
|----------------------------|------------------------|
| GET /api/knowledge         | List knowledge entries |
| POST /api/knowledge        | Create entry           |
| PUT /api/knowledge/{id}    | Update entry           |
| DELETE /api/knowledge/{id} | Delete entry           |

## Tasks

| Endpoint               | Description |
|------------------------|-------------|
| GET /api/tasks         | List tasks  |
| POST /api/tasks        | Create task |
| PUT /api/tasks/{id}    | Update task |
| DELETE /api/tasks/{id} | Delete task |

## Real-Time

| Endpoint   | Description                                       |
|------------|---------------------------------------------------|
| WS /api/ws | WebSocket - receives real-time data change events |

# 7. Database Schema

The database uses 10+ tables with foreign key relationships. All tables use auto-increment integer primary keys and track creation/update timestamps.

## Tables

| Table       | Description             | Key Fields                                                                           |
|-------------|-------------------------|--------------------------------------------------------------------------------------|
| users       | User accounts & auth    | email, password_hash, role, name                                                     |
| companies   | Business entities (CRM) | name, phone, email, address, industry, size                                          |
| contacts    | Contact persons         | name, phone, email, position, company_id (FK)                                        |
| projects    | Tracked projects        | name, status, budget, company_id (FK)                                                |
| leads       | Sales leads pipeline    | name, status (NEW/QUALIFIED/CONTACTED/NEGOTIATING/WON/LOST), source, company_id (FK) |
| quotes      | Price quotations        | quote_number, total, status, lead_id (FK), company_id (FK)                           |
| activities  | Audit log               | entity_type, entity_id, action, description, created_by (FK users)                   |
| knowledge   | Internal knowledge base | title, content, category, tags                                                       |
| tasks       | Assignable tasks        | title, description, status, assigned_to (FK users), due_date                         |
| quote_items | Line items in quotes    | quote_id (FK), description, quantity, unit_price                                     |

## 8. Real-Time WebSocket System

MED.MIX OS includes a built-in WebSocket server for real-time multi-user synchronization. When any user creates, updates, or deletes a record, a broadcast message is sent to all connected clients, triggering an automatic data refresh.

### Architecture

The WebSocket system is implemented in `apps/backend/app/services/realtime.py`. It uses a simple in-memory connection manager with a broadcast function.

### Connection

```
// Connect from browser
const ws = new WebSocket('wss://backend-production-40db.up.railway.app/ws');
ws.onmessage = (event) => {
  const msg = JSON.parse(event.data);
  if (msg.event === 'data_changed') {
    refreshData();
  }
};
```

### Event Format

| Field       | Type    | Description                               |
|-------------|---------|-------------------------------------------|
| event       | string  | Always 'data_changed'                     |
| entity_type | string  | Affected entity (company, lead, etc.)     |
| entity_id   | integer | Affected record ID                        |
| action      | string  | Action performed (create, update, delete) |

### Frontend Integration

The frontend uses a custom `useRealtime` hook:

```
import { useRealtime } from '@lib/realtime';

function MyPage() {
  const [data, setData] = useState(null);
  const refresh = useCallback(() => fetchData().then(setData), []);
  useRealtime(refresh); // Auto-refresh on any change
  return ...;
}
```

### Auto-Reconnect

The WebSocket client automatically reconnects every 3 seconds if the connection is lost. This ensures resilience against temporary network interruptions and Railway deployments.

## 9. AI Assistant

The AI Assistant provides context-aware business intelligence powered by OpenAI's GPT-4o-mini model. It loads CRM context (companies, contacts, active leads) and answers questions about the business data.

### How It Works

- User sends a message via the chat interface
- Backend loads CRM context: company/contact/lead summaries
- System prompt + user message sent to OpenAI API
- OpenAI returns a natural language response
- Response is streamed/sent back to the user

### Fallback Mode

If the OpenAI API key is not configured, the assistant falls back to simple keyword-based response matching for basic queries.

### Configuration

```
# Required environment variable
OPENAI_API_KEY=sk-proj-...

# Model: gpt-4o-mini (cheapest, $0.15/1M input tokens)
# Free tier: $5 credit covers ~33,000 conversations
```

### API Endpoint

| Endpoint                 | Usage                                |
|--------------------------|--------------------------------------|
| POST /api/assistant/chat | Send message body={'message': '...'} |

**Note:** The AI assistant only has read-only access to CRM data. It cannot create, update, or delete records.

## 10. OpenStreetMap Business Search

MED.MIX OS includes a business data import feature powered by OpenStreetMap's Overpass API. This allows users to search for companies by type and location, completely free and without any API key.

### How It Works

- User specifies business type (e.g., 'restaurant', 'pharmacy', 'school')
- User specifies location (city or coordinates)
- Backend queries OpenStreetMap Overpass API for matching nodes/ways
- Results are parsed into lead entries in the system
- Optional: Google Maps fallback if GOOGLE\_MAPS\_API\_KEY is configured

### API Endpoint

| Endpoint                      | Usage                                               |
|-------------------------------|-----------------------------------------------------|
| POST /api/leads/import/search | Body: {'query': 'restaurant', 'location': 'Riyadh'} |

### Example Request

```
POST /api/leads/import/search
Content-Type: application/json
Authorization: Bearer <token>

{
  "query": "pharmacy",
  "location": "Jeddah"
}
```

### Rate Limits

OpenStreetMap Overpass API has a fair use policy. For normal usage (a few searches per minute), there are no practical limits. For bulk imports, consider spacing requests 2-3 seconds apart.

# 11. File Storage

MED.MIX OS uses MinIO (S3-compatible object storage) for file uploads. In production on Railway, files are stored on Railway Volumes instead.

## Configuration

```
# Environment variables for file storage
MINIO_ENDPOINT=localhost:9000
MINIO_ACCESS_KEY=minioadmin
MINIO_SECRET_KEY=minioadmin
MINIO_BUCKET=medmix-files

# For Railway: use local volume storage
# Mount a Railway Volume at /app/uploads
```

## Supported File Types

Images (JPEG, PNG, GIF, WebP), Documents (PDF, DOCX, XLSX), and other common business file types. File size limit: 10MB per upload.

## 12. Email System

MED.MIX OS includes an email notification system that requires SMTP credentials to function. Currently this is an optional feature.

### Configuration

```
# SMTP settings (optional)
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_USER=your-email@gmail.com
SMTP_PASSWORD=your-app-password
SMTP_FROM=noreply@medmix.com
```

**Note:** For Gmail: Use an App Password (not your regular password). Enable 2-Step Verification first, then generate an App Password from Google Account settings.

### Email Features (Planned)

- Welcome email on registration
- Password reset via email
- Lead assignment notifications
- Task deadline reminders
- Quote approval requests

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## 14. Environment Variables

| Variable            | Required | Description                       |
|---------------------|----------|-----------------------------------|
| DATABASE_URL        | Yes      | PostgreSQL connection string      |
| SECRET_KEY          | Yes      | JWT signing secret (min 32 chars) |
| OPENAI_API_KEY      | No       | OpenAI API key for AI assistant   |
| SMTP_HOST           | No       | SMTP server for email             |
| SMTP_PORT           | No       | SMTP port (usually 587)           |
| SMTP_USER           | No       | SMTP username/email               |
| SMTP_PASSWORD       | No       | SMTP password or app password     |
| SMTP_FROM           | No       | Sender email address              |
| GOOGLE_MAPS_API_KEY | No       | Optional Google Maps fallback     |
| MINIO_ENDPOINT      | No       | MinIO server endpoint             |
| MINIO_ACCESS_KEY    | No       | MinIO access key                  |
| MINIO_SECRET_KEY    | No       | MinIO secret key                  |
| PORT                | No       | Server port (default: 8000)       |
| NEXT_PUBLIC_API_URL | No       | Frontend: API base URL            |

# 15. Troubleshooting

## Backend Health Check

```
curl https://backend-production-40db.up.railway.app/api/health
# Expected: {"status":"ok", "app":"MED.MIX OS"}
```

## Common Issues

| Issue                       | Solution                                                                   |
|-----------------------------|----------------------------------------------------------------------------|
| Backend returns 502         | Wait 30s for Railway to finish deploying; check health endpoint            |
| Database connection failed  | Verify DATABASE_URL uses postgresql+asyncpg:// scheme                      |
| Login returns 401           | Check username/password; ensure user is registered                         |
| CORS error on frontend      | Verify NEXT_PUBLIC_API_URL matches backend URL; CORS configured in main.py |
| WebSocket not connecting    | Ensure URL uses wss:// protocol; check Railway firewall                    |
| AI Assistant not responding | Verify OPENAI_API_KEY is set and has available credits                     |
| Search returns no results   | Try broader query term; verify location name in English                    |
| File upload fails           | Check MinIO or Railway volume configuration                                |

## Getting Help

GitHub Repository: <https://github.com/Fly89by/medmix>

Report issues: <https://github.com/Fly89by/medmix/issues>